Harish G

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 2 years of experience specializing in firmware development and real-time systems. Proven ability to deliver reliable embedded solutions and a strong commitment to continuous learning and professional growth.

SKILLS

Languages	:	C, C++, Embedded C, Linux System Programming
Operating Systems	:	Linux, Windows, RTOS (porting to Linux)
Embedded Hardware	:	ARM7, SAMA5D3, NMC, LPC2129
Communication Protocols	:	UART, RS485, I2C, SPI, CAN, SLIP
Networking Protocols	:	HTTP/HTTPS, REST API, SSH/CLI, Telnet, FTPS, SNMP, SCP, TCP/IP
Libraries	:	cJSON, Jansson, libcurl, OpenSSL, zlib, WPA Client
Core Concepts	:	Linux IPC (Pipe, FIFO, Message Queue, Shared Memory, Semaphore, Mutex), Multithreading (pthreads), Socket Programming, State Machines
Development Tools	:	Git, GitKraken, Eclipse, VS Code, Jira
Project Tools	:	Postman, PuTTY, MIB Browser, Tera term, Wireshark, RDP, WSL, FileZilla

WORK EXPERIENCE

nVent Electrical India Pvt Ltd, Bangalore Feb 2025 – Present

Embedded Software Engineer

Enlogic Power Distribution Unit

- Worked as part of the Firmware Development team at nVent, developing and maintaining firmware for intelligent Power Distribution Units (PDUs) with multiple management interfaces including Web UI, SNMP, Redfish, CLI, SSH, and Telnet. The PDUs are designed for data centers, offering advanced monitoring and remote management capabilities.
- Hands-on with libraries: cJSON, Jansson, libcurl, OpenSSL, zlib, WPA Client. Configured nVent products such as Enlogic, Rack Power, and Dell IPv6 PDUs. Integrated sensors: temperature, humidity, door, smoke, dry contact, and spot fluid.
- Performed issue triage, root cause analysis, bug fixing, and feature implementation, ensuring system reliability and performance.
- Conducted developer-level testing, regression verification, and code reviews for continuous improvement.
- Worked with RESTful APIs and system level APIs to enhance device integration and remote management.

Bi-directional Serial–TCP Bridge for Remote Device Access

- Designed and implemented a C/Linux application to bridge physical serial interfaces (ttyS0) with TCP sockets, enabling remote device access without physical serial cables.
- Developed TCP client/server modules and integrated with PTY master/slave, getty, and system calls to emulate serial communication over the network.
- Utilized fork, sockets, and system calls to replicate minicom-like functionality across multiple devices with pre-installed libraries.
- Eliminated the need for engineers to be physically present at device racks, improving efficiency and reducing downtime.

OOLC – Outlet Overload Control (Cisco - Enlogic PDU Feature)

- Built per-outlet overload protection in C/Linux: 80% warning and 100% overload stages with debounce, auto-shutdown, and latched-off state until manual restore.
- Persisted settings and shutdown state in EEPROM; added event logs, SNMP traps, alarm-page updates, and privilege-checked enable/disable with audit trail.
- Extended OOLC to daisy-chained PDUs: master polls slave alarms and logs every stage transition, even when polls skip intermediate stages.

SCHNEIDER ELECTRIC P– RTOS to Linux State Machine Porting

- Ported the RS485/SLIP receive state machine (IDLE → START → DATA → COMPLETE/ERROR) from RTOS to Linux, replacing RTOS semaphores with POSIX UART reads and timeouts.
- Fixed the outlet-drop issue: ACK/NACK validation, bus-collision purge, start-byte resync, retry tuning, and rejection of corrupted all-zero voltage reads.
- Added timing-based quiet-period and CRC checks, keeping outlet and breaker status reads stable on the shared RS485 bus.

Vector India, Bangalore Nov 2024 – Feb 2025

Embedded Lab Coordinator

- Guided newly enrolled trainees in C, C++, Embedded C, and Linux IPC concepts during scheduled lab sessions.
- Provided hands-on support with debugging, syntax correction, and logic building for embedded programming exercises.

Vector India, Bangalore Feb 2024 – Sep 2024

Embedded Systems Trainee

- Completed intensive hands-on training in Embedded C, C++, Data Structures, and Microcontroller Programming.
- Worked on real-time projects involving UART, SPI, I2C, and CAN protocols.
- Gained practical experience in firmware debugging, interrupt handling, and memory management.
- Implemented Linux IPC mechanisms including Pipe, FIFO, Message queue, Shared memory, Semaphore, Mutex and threads.

PROJECT WORK

Employee Attendance System Using RFID and LINUX

- Developed the attendance system using LPC2129 microcontroller, RFID, UART, and I2C.
- Leveraged Radio Frequency Identification (RFID) technology to record attendance data, with a Linux-based server for data management, storage, and processing.
- Used an I2C-based Real Time Clock (RTC) module for accurate timekeeping and UART to transmit data from the microcontroller to a Linux monitor.

Student Database Management System (Data Structures using C++)

- Designed a system to manage student records using a singly linked list for efficient data handling.
- Maintained student records by roll number, name, and marks, and used the file system to store the data.
- Features included adding, viewing, searching, modifying, sorting, and deleting records.

EDUCATION

Government College of Engineering, Erode Aug 2019 – Jun 2023

B.E. in Electronics and Communication Engineering ***CGPA: 7.5/10***

Focused on core electronics and programming fundamentals with exposure to embedded systems and software development.

TRAINING AND CERTIFICATES

- Completed Advanced Embedded System Course at Vector India Pvt. Ltd, Bangalore.
- Lab Coordinator at Vector India Pvt. Ltd, Bangalore.